DISTRIBUCIONES CONTINUAS

$f(x)$: densidad $\quad F(x) = P(X \le x) \quad M(t) = E[e^{tX}]$.

$M(0) = 1$ siempre. Fuera del dominio indicado de M, la esperanza exponencial es $+\infty$. Tasa λ y escala β: $\beta = 1/\lambda$.

Uniforme $a < b$

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{otro caso.} \end{cases} \qquad F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x < b, \\ 1, & x \ge b. \end{cases}$$

$$M(t) = \begin{cases} \dfrac{e^{bt} - e^{at}}{(b-a)t}, & t \neq 0, \\ 1, & t = 0. \end{cases} \qquad t \in \mathbb{R}.$$

Normal $\mu \in \mathbb{R},\ \sigma > 0$

$$X \sim \mathcal{N}(\mu, \sigma^2), \qquad x \in \mathbb{R},$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

$$M(t) = e^{\mu t + \sigma^2 t^2/2}, \qquad t \in \mathbb{R}.$$

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2}\, du.$$

Exponencial tasa $\lambda > 0$; escala $\beta > 0$

$$f(x) = \begin{cases} \lambda e^{-\lambda x} = \dfrac{1}{\beta} e^{-x/\beta}, & x \ge 0, \\ 0, & x < 0. \end{cases}$$

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x} = 1 - e^{-x/\beta}, & x \ge 0. \end{cases}$$

$$M(t) = \frac{\lambda}{\lambda - t} = \frac{1}{1 - \beta t}, \qquad t < \lambda = \frac{1}{\beta}.$$

Gamma / Erlang forma $\alpha > 0$; $\beta = 1/\lambda$

$$\Gamma_{\text{tasa}}(\alpha, \lambda) = \Gamma_{\text{escala}}(\alpha, \beta).$$

$$f(x) = \begin{cases} \dfrac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} = \dfrac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

$$F(x) = \begin{cases} 0, & x \le 0, \\ P(\alpha, \lambda x) = P(\alpha, x/\beta), & x > 0. \end{cases}$$

$$M(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha = (1 - \beta t)^{-\alpha}, \quad t < \lambda = \frac{1}{\beta}.$$

Erlang: $\alpha = r \in \mathbb{N}$, $x \ge 0$:

$$F(x) = 1 - e^{-\lambda x} \sum_{j=0}^{r-1} \frac{(\lambda x)^j}{j!}.$$

Chi-cuadrada $\nu > 0$

$$\chi^2_\nu = \Gamma_{\text{tasa}}(\nu/2, 1/2) = \Gamma_{\text{escala}}(\nu/2, 2).$$

$$f(x) = \begin{cases} \dfrac{x^{\nu/2-1} e^{-x/2}}{2^{\nu/2}\Gamma(\nu/2)}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

$$F(x) = \begin{cases} 0, & x \le 0, \\ P(\nu/2, x/2), & x > 0, \end{cases} \qquad M(t) = (1-2t)^{-\nu/2}, \quad t < \tfrac{1}{2}.$$

Beta $\alpha, \beta > 0$: ambos de forma

$$f(x) = \begin{cases} \dfrac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, & 0 < x < 1, \\ 0, & \text{otro caso.} \end{cases}$$

$$F(x) = \begin{cases} 0, & x \le 0, \\ I_x(\alpha, \beta), & 0 < x < 1, \\ 1, & x \ge 1. \end{cases}$$

$$M(t) = \sum_{j=0}^{\infty} \frac{(\alpha)_j}{(\alpha+\beta)_j} \frac{t^j}{j!}, \qquad t \in \mathbb{R}.$$

Cauchy localización $x_0 \in \mathbb{R}$; escala $\gamma > 0$

$$f(x) = \frac{1}{\pi\gamma[1 + ((x - x_0)/\gamma)^2]}, \quad x \in \mathbb{R},$$

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x - x_0}{\gamma}\right).$$

$$M(t) = \begin{cases} 1, & t = 0, \\ +\infty, & t \neq 0. \end{cases} \qquad \text{No existe FGM alrededor de 0.}$$

Laplace localización $\mu \in \mathbb{R}$; escala $b = 1/\lambda > 0$

$$f(x) = \frac{1}{2b} e^{-|x-\mu|/b} = \frac{\lambda}{2} e^{-\lambda|x-\mu|}, \quad x \in \mathbb{R}.$$

$$F(x) = \begin{cases} \dfrac{1}{2} e^{(x-\mu)/b} = \dfrac{1}{2} e^{\lambda(x-\mu)}, & x < \mu, \\ 1 - \dfrac{1}{2} e^{-(x-\mu)/b} = 1 - \dfrac{1}{2} e^{-\lambda(x-\mu)}, & x \ge \mu. \end{cases}$$

$$M(t) = \frac{e^{\mu t}}{1 - b^2 t^2} = \frac{e^{\mu t} \lambda^2}{\lambda^2 - t^2}, \qquad |t| < 1/b = \lambda.$$

Notación auxiliar

$$\Gamma(a) = \int_0^\infty u^{a-1} e^{-u}\, du, \qquad \Gamma(n) = (n-1)!,$$

$$P(a, z) = \frac{1}{\Gamma(a)} \int_0^z u^{a-1} e^{-u}\, du, \qquad B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

$$I_x(a,b) = \frac{1}{B(a,b)} \int_0^x u^{a-1}(1-u)^{b-1}\, du.$$

$(a)_0 = 1$; $\quad (a)_j = a(a+1)\cdots(a+j-1)$.

DISTRIBUCIONES DISCRETAS

$p(k) = P(X = k) \quad F(x) = P(X \le x) \quad M(t) = E[e^{tX}].$

$q = 1 - p$, $0 < p < 1$; $\mathbb{N}_0 = \{0, 1, \ldots\}$; $\lfloor x \rfloor$: parte entera inferior. Suma vacía $= 0$; $M(0) = 1$; fuera del dominio de M, $+\infty$.

Bernoulli

$$p(k) = \begin{cases} q, & k = 0, \\ p, & k = 1, \\ 0, & \text{otro caso}, \end{cases} \qquad F(x) = \begin{cases} 0, & x < 0, \\ q, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

$$M(t) = q + pe^t, \qquad t \in \mathbb{R}.$$

Binomial $n \in \mathbb{N}$

$$p(k) = \begin{cases} \binom{n}{k} p^k q^{n-k}, & k = 0, \ldots, n, \\ 0, & \text{otro caso}. \end{cases}$$

$$F(x) = \begin{cases} 0, & x < 0, \\ \displaystyle\sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k q^{n-k}, & 0 \le x < n, \\ 1, & x \ge n. \end{cases}$$

$$M(t) = (q + pe^t)^n, \qquad t \in \mathbb{R}.$$

Poisson $\theta > 0$; $\theta = \lambda h$ en un intervalo h

$$p(k) = \begin{cases} e^{-\theta} \dfrac{\theta^k}{k!}, & k \in \mathbb{N}_0, \\ 0, & \text{otro caso}, \end{cases}$$

$$F(x) = \begin{cases} 0, & x < 0, \\ e^{-\theta} \displaystyle\sum_{k=0}^{\lfloor x \rfloor} \frac{\theta^k}{k!}, & x \ge 0. \end{cases} \qquad M(t) = e^{\theta(e^t - 1)}, \quad t \in \mathbb{R}.$$

Uniforme discreta $a, b \in \mathbb{Z},\ a \le b$; $m = b - a + 1$

$$p(k) = \begin{cases} \dfrac{1}{m}, & k = a, \ldots, b, \\ 0, & \text{otro caso}, \end{cases} \qquad F(x) = \begin{cases} 0, & x < a, \\ \dfrac{\lfloor x \rfloor - a + 1}{m}, & a \le x < b, \\ 1, & x \ge b. \end{cases}$$

$$M(t) = \begin{cases} \dfrac{e^{at}(e^{mt} - 1)}{m(e^t - 1)}, & t \ne 0, \\ 1, & t = 0, \end{cases} \qquad t \in \mathbb{R}.$$

Hipergeométrica $N \ge 2,\ 0 \le K, n \le N$ enteros
$L = \max(0, n - N + K), \quad U = \min(n, K)$.

$$p(k) = \begin{cases} \dfrac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, & k = L, \ldots, U, \\ 0, & \text{otro caso}. \end{cases}$$

$$F(x) = \begin{cases} 0, & x < L, \\ \displaystyle\sum_{k=L}^{\lfloor x \rfloor} p(k), & L \le x < U, \\ 1, & x \ge U. \end{cases}$$

$$M(t) = \sum_{k=L}^{U} e^{tk} \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \qquad t \in \mathbb{R}.$$

Geométrica ensayos T; fracasos $Y = T - 1$

$$p_T(k) = \begin{cases} pq^{k-1}, & k \in \mathbb{N}, \\ 0, & \text{otro caso}, \end{cases} \qquad p_Y(k) = \begin{cases} pq^k, & k \in \mathbb{N}_0, \\ 0, & \text{otro caso}. \end{cases}$$

$$F_T(x) = \begin{cases} 0, & x < 1, \\ 1 - q^{\lfloor x \rfloor}, & x \ge 1, \end{cases} \qquad F_Y(x) = \begin{cases} 0, & x < 0, \\ 1 - q^{\lfloor x \rfloor + 1}, & x \ge 0. \end{cases}$$

$$M_T(t) = \frac{pe^t}{1 - qe^t}, \qquad M_Y(t) = \frac{p}{1 - qe^t}, \qquad t < -\ln q.$$

Binomial negativa $r \in \mathbb{N}$ éxitos
Ensayos T; fracasos $Y = T - r$.

$$p_T(k) = \begin{cases} \binom{k-1}{r-1} p^r q^{k-r}, & k = r, r+1, \ldots, \\ 0, & \text{otro caso}, \end{cases}$$

$$p_Y(k) = \begin{cases} \binom{k+r-1}{r-1} p^r q^k, & k \in \mathbb{N}_0, \\ 0, & \text{otro caso}. \end{cases}$$

$$F_T(x) = \begin{cases} 0, & x < r, \\ \displaystyle\sum_{k=r}^{\lfloor x \rfloor} p_T(k), & x \ge r, \end{cases} \qquad F_Y(x) = \begin{cases} 0, & x < 0, \\ \displaystyle\sum_{k=0}^{\lfloor x \rfloor} p_Y(k), & x \ge 0. \end{cases}$$

$$M_T(t) = \left(\frac{pe^t}{1 - qe^t}\right)^r, \quad M_Y(t) = \left(\frac{p}{1 - qe^t}\right)^r, \quad t < -\ln q.$$

Multinomial $n \in \mathbb{N},\ p_i > 0,\ \sum_i p_i = 1$
$\mathcal{S} = \{\mathbf{k} \in \mathbb{N}_0^d : \sum_i k_i = n\}$.

$$p(\mathbf{k}) = \begin{cases} \dfrac{n!}{\prod_i k_i!} \prod_{i=1}^d p_i^{k_i}, & \mathbf{k} \in \mathcal{S}, \\ 0, & \text{otro caso}. \end{cases}$$

$$F(\mathbf{x}) = \sum_{\substack{\mathbf{k} \in \mathcal{S} \\ k_i \le x_i\ \forall i}} p(\mathbf{k}), \qquad M(\mathbf{t}) = \left(\sum_{i=1}^d p_i e^{t_i}\right)^n, \quad \mathbf{t} \in \mathbb{R}^d.$$

Hipergeométrica multivariada $N = \sum_i N_i$
$N_i \in \mathbb{N}$, $0 \le n \le N$ entero, $\mathcal{H} = \{\mathbf{k} \in \mathbb{N}_0^d : k_i \le N_i,\ \sum_i k_i = n\}$.

$$p(\mathbf{k}) = \begin{cases} \dfrac{\prod_{i=1}^d \binom{N_i}{k_i}}{\binom{N}{n}}, & \mathbf{k} \in \mathcal{H}, \\ 0, & \text{otro caso}. \end{cases}$$

$$F(\mathbf{x}) = \sum_{\substack{\mathbf{k} \in \mathcal{H} \\ k_i \le x_i\ \forall i}} p(\mathbf{k}), \qquad M(\mathbf{t}) = \sum_{\mathbf{k} \in \mathcal{H}} e^{\sum_i t_i k_i} p(\mathbf{k}), \quad \mathbf{t} \in \mathbb{R}^d.$$